Assignment 6: CNN Classification on EuroSAT (Transfer Learning)

Applied Machine Learning for Software and Computer Engineering (SEG4180)
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Dataset and Preprocessing

The EuroSAT_RGB dataset from Hugging Face was used for land-use classification. It contains 27,000 RGB satellite images (64×64 pixels) across 10 balanced classes, including *Forest, River, Residential Buildings, Industrial Buildings*, and *SeaLake*.

Preprocessing steps:

- Images resized to 224×224 pixels.
- MobileNetV2 `preprocess_input` normalization applied.
- Official train/validation/test splits used (16,200 / 5,400 / 5,400).

Model Architecture

Transfer learning was applied using MobileNetV2 pretrained on ImageNet. The convolutional backbone was first frozen to leverage pretrained feature extraction, followed by a classification head consisting of Global Average Pooling, a Dense layer with 128 ReLU units, Dropout (0.3), and a final Dense softmax layer with 10 outputs. Training was performed in two stages: an initial frozen-backbone phase using Adam (lr=1e-3), followed by fine-tuning of the last 20% of layers with a reduced learning rate (1e-5). EarlyStopping and ReduceLROnPlateau were employed to stabilize convergence.

Results

Test Accuracy: 94.93% & Macro F1-score: 0.9476

The model achieved strong and consistent performance across all classes, as reflected by the close alignment between macro and weighted F1-scores Highest performance was observed in *Residential Buildings* and *SeaLake*, likely due to their distinctive spatial patterns. Lower performance occurred in *River* and *Highway*, possibly due to visual similarity with other linear or water-related categories. Validation curves indicate stable convergence with no significant overfitting observed.

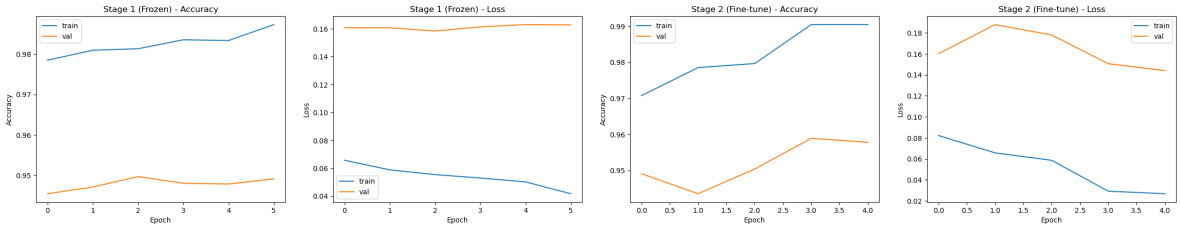

Figure 1: Training and Validation Accuracy/Loss Across Both Training Stages

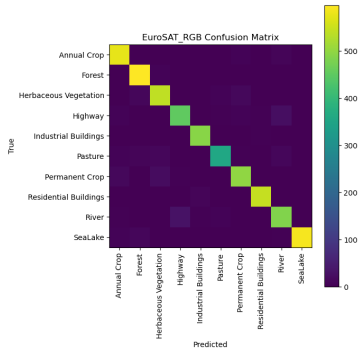

Figure 2: Confusion Matrix on Test Set

Discussion

The two-stage transfer learning strategy enabled rapid convergence while maintaining strong generalization. Freezing the backbone allowed efficient feature reuse, while fine-tuning improved class-level discrimination. Performance variations among visually similar classes highlight opportunities for further gains through enhanced data augmentation or deeper fine-tuning.